

GMD AI-Assisted Harmonization System

Infrastructure Request for ITS Review

GPID Team

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PURPOSE

This note summarizes the infrastructure required for a proof of concept of the GMD AI-assisted harmonization pipeline. It is intended to support ITS review and to obtain clear guidance on which services, access patterns, and security controls are available in the WBG Azure environment.

1 Summary

The GPID team is developing an AI-assisted workflow to support harmonization of household surveys into the Global Monitoring Database.

The operating model is straightforward:

- The system reads survey documentation and drafts a structured Harmonization Specification.
- A GMD expert reviews and approves the specification.
- A deterministic execution engine runs only approved specifications.

This means AI supports drafting, while final methodological decisions and production execution remain under human control.

2 What We Need from ITS

We are asking ITS to confirm the availability and configuration of the infrastructure required to operate this workflow in a policy-compliant Azure environment. In particular, we need guidance on:

- approved storage for restricted survey data and documentation;
- approved database services;
- approved access to enterprise AI services;
- approved services for similarity-based retrieval of historical decisions;
- an approved compute environment for a team-managed Python pipeline; and
- standard logging and monitoring services, with direct read access for the GPID team.

3 Division of Responsibilities

Table 1: Division of Responsibilities

ITS Responsibilities	GPID Team Responsibilities
Provision and secure the infrastructure	Develop and maintain the pipeline code
Approve the relevant Azure services and access model	Design prompts, rules, and harmonization logic
Ensure compliance with WBG security and data policies	Review and approve all AI-generated specifications
Maintain the underlying platform services	Operate the harmonization workflow within the approved environment

The GPID team will need direct access to deploy code, monitor jobs, and review logs within the approved environment. The workflow is not practical if each deployment or troubleshooting step requires a separate ITS ticket.

4 Required Infrastructure Components

4.1 Storage

We need secure cloud storage for survey microdata, documentation, legacy harmonization scripts, and pipeline outputs. This storage must support controlled access by the pipeline and preserve original files for audit and reproducibility.

We expect Azure Blob Storage to be the most likely fit, subject to ITS guidance on approved configurations for restricted data.

Where available, we intend to ingest DDI metadata from the World Bank Microdata Library API at <https://microdata.worldbank.org>. This requires outbound access from the compute environment to that service.

ITS QUESTION 1

Is there an approved Azure Blob Storage environment where we can store restricted household survey microdata? If so, what access controls and data classification requirements apply?

ITS QUESTION 2

Can our compute environment make outbound API calls to `microdata.worldbank.org` to retrieve DDI metadata files?

4.2 Relational Database

We need a relational database to store the canonical definitions of GMD variables, validation rules, dependencies, and other structured metadata used by the pipeline.

This database is the authoritative source for variable definitions. Historical precedents may inform a recommendation, but they must not override the canonical schema.

Azure SQL Database or PostgreSQL on Azure would both satisfy the requirement. We defer to ITS on the approved service in our environment.

ITS QUESTION 3

What is the standard approved relational database service in the WBG Azure environment: Azure SQL, PostgreSQL, or another option? Is there an existing instance we could use, or would a new one need to be provisioned?

4.3 Enterprise AI Service

We need an enterprise-approved large language model service to draft Harmonization Specifications from survey documentation and reference materials.

Our preferred option is Azure OpenAI Service, subject to WBG approval and data classification rules. This is the key dependency for the AI design.

The AI will primarily process survey documentation and metadata rather than respondent-level microdata, but we need ITS guidance on what data categories are permitted.

If Azure OpenAI is not approved, we need the approved alternative for running large language models within the WBG environment.

! ITS QUESTION 4 – PRIORITY

Is Azure OpenAI Service approved for use in the WBG Azure tenant? If yes, what data classification levels are permitted, and are there restrictions on what data may be sent to it? If no, what is the approved path for using large language models within the WBG environment?

4.4 Historical Decisions Retrieval

We need a searchable store of past harmonization decisions so the system can retrieve relevant precedents when drafting new specifications.

This is distinct from the rules database. The rules database is authoritative. The historical store is advisory and is intended to improve consistency and reuse institutional knowledge.

The retrieval requirement is similarity-based rather than exact-match only, so we expect a vector-capable search service to be required.

Azure AI Search with vector search appears to be the most likely fit, subject to ITS guidance.

💡 ITS QUESTION 5

Is Azure AI Search available and approved in our environment with vector search capabilities enabled? If not, is there an alternative approved service for similarity-based document retrieval?

4.5 Compute Environment

We need an approved compute environment to run a Python-based pipeline that connects to storage, databases, the AI service, and the historical retrieval service.

Our target production architecture is Azure Kubernetes Service (AKS). For the proof of concept, we are open to a simpler approved option if that accelerates setup, provided it does not block later migration to AKS.

For the GPID team to operate effectively, the environment must allow:

- self-service code deployment;
- direct read access to job and application logs; and
- a separate non-production environment for development and testing.

 ITS QUESTION 6

Is Azure Kubernetes Service (AKS) available for a team-managed workload, and can our team have self-service deployment and log access?

ITS QUESTION 7

If AKS is not immediately available, what is the simplest approved compute environment for running scheduled Python-based jobs? Can ITS confirm that AKS remains the path for production?

4.6 Logging, Monitoring, and Audit

We need standard platform logging and monitoring for pipeline jobs, together with a structured audit trail of inputs, approvals, execution versions, outputs, and validation results.

We expect detailed logs to flow to the standard Azure monitoring service used by ITS, while structured run records can be stored in the relational database.

 ITS QUESTION 8

What is the standard logging and monitoring service in our Azure environment? Can our jobs be onboarded to it? What is the retention policy? Can our team query logs directly without routing requests through ITS?

5 Access Model Required for the PoC

For the proof of concept to be workable, the GPID team needs:

- the ability to deploy updated code without filing an ITS request for each change;
- direct visibility into pipeline logs for debugging and operational review; and
- a development environment separated from any production or restricted-data environment.

This access model is a requirement for the PoC, not a later enhancement.

6 Scope of the Proof of Concept

The proof of concept covers non-welfare demographic variables only, including sex, age, education, urban or rural classification, household relationships, and labor force status. Welfare aggregates, including income and consumption, are out of scope.

Expected annual throughput is approximately 100 to 200 new surveys. The PoC will begin with a smaller representative subset.

7 Consolidated Questions for ITS

Table 2: Consolidated ITS Questions

#	Topic	Question
Q1	Storage	Is there an approved Azure Blob Storage environment for restricted household survey microdata? What access controls and data classification requirements apply?
Q2	Storage / API	Can our compute environment make outbound API calls to <code>microdata.worldbank.org</code> to retrieve DDI metadata files?
Q3	Relational database	What is the standard approved relational database service in the WBG Azure environment, and is an existing instance available?
Q4	AI service	Is Azure OpenAI Service approved in the WBG Azure tenant? If yes, what data classification levels are permitted? If no, what is the approved alternative for using large language models?
Q5	Historical retrieval	Is Azure AI Search available and approved with vector search capabilities? If not, what approved alternative is available?
Q6	Compute	Is AKS available for a team-managed workload, and can our team have self-service deployment and log access?
Q7	Compute / PoC path	If AKS is not immediately available, what simpler approved compute environment should be used for the PoC, and can ITS confirm AKS remains the production path?
Q8	Monitoring	What is the standard logging and monitoring service, what is the retention policy, and can our team query logs directly?

8 What We Are Not Asking ITS to Build

We are not asking ITS to design the harmonization methodology, develop the AI workflow, write prompts, or implement the pipeline logic.

We are asking ITS to confirm and provision the infrastructure, security model, and access pattern required for the GPID team to build and operate the solution within the approved environment.